

Review of Related Literature

According to Katarzyna Bernat (2023), effective post-consumer plastic waste management is crucial to mitigate the environmental impacts of plastic pollution. The challenges associated with plastic waste management range from littering to high collection costs and low recycling rates. To address these issues, several countries have transformed their collection systems, but collecting more plastic packaging comes at a cost, as the feedstock for the sorting process becomes more complex and leads to cross-contamination within the sorted fractions. Therefore, a balance must be obtained between packaging design, collection and recycling rates, and the quality of fractions that have been sorted.

The development of advanced technologies, such as automated waste collection monitoring systems and multi-sensory artificial intelligence (AI) for sorting plastic waste, can significantly improve the efficiency of recycling plastic waste. Additionally, blockchain technology can be used to create a trust-based platform between plastic waste separators, recyclers, and buyers of recycled raw materials, enabling accurate tracking and payment for recycled materials.

The study highlights the importance of separate collection systems for recyclable plastics and the need for further investment in pretreatment, sorting, and recycling technologies. Simpler recyclable packaging designs can also contribute to increased plastic recycling rates. Moreover, the use of digital product passports can provide more information about ingredients, enabling higher-value new products from municipal waste.

The review emphasizes the need for more data on the type of containers and plastics, as well as the frequency of unsorted waste collection. By leveraging advanced technologies and innovative approaches, the recycling industry can improve the quality and quantity of recycled plastics, reducing the environmental impacts of plastic waste and promoting a more circular economy. <https://doi.org/10.3390/en16083504>

According to Davide Anghinolfi et al. (2012), a dynamic optimization model for solid waste recycling can significantly improve waste management efficiency. The model focuses on recycling management and dynamic optimization of materials collection, minimizing costs and environmental impact. By integrating GIS and optimization tools, the model provides a decision support system for waste collection and recycling operations.

The study highlights the importance of dynamic optimization in waste management, allowing for real-time adjustments based on waste generation patterns and bin levels. The model can be used to optimize waste collection routes, reducing costs and environmental impact. In a case study of the Cogoleto municipality, the optimized collection resulted in net benefits approximately 2.5 times greater than the estimated current policy. <https://doi.org/10.1016/j.wasman.2012.10.006>

According to Juan Valenzuela et al. (2020), a literature review of reverse logistics models for collecting plastic waste highlights the importance of circular economy models and RL networks in promoting sustainability.

The study reviews articles published between 2014 and 2019 and finds that mixed-integer linear programming is the most widely used solution method for mathematical modeling, while evaluations are the most common approach for exploratory studies. <https://doi.org/10.1080/16258312.2008.11517186>

According to Iva Kovacic et al. (2020), the construction industry is one of the world's largest consumers of resources and produces significant amounts of waste.

The study proposes a framework for a Digital Platform for Circular Economy (DEEP) that integrates various stakeholders and data repositories to optimize the use of resources in the circular economy. <https://doi.org/10.25219/epoj.2020.00107>

According to Linda Godfrey and Suzan Oelofse (2017), South Africa's waste management and recycling sector has undergone significant changes over the past three decades, driven by social and economic needs. The country has a local recycling economy comparable to many developed countries, thanks in part to a large and active informal waste sector.

The study highlights the importance of the informal waste sector in South Africa, which is responsible for collecting 80-90% of post-consumer paper and packaging recyclables. The authors suggest that South Africa is standing on the brink of a new phase, "The future is a Circular Economy," and that the country has the opportunity to leapfrog developed countries' technology choices and move directly towards a stronger materials recovery paradigm. <https://doi.org/10.3390/resources6040057>